Problem 7.13

Derive a formula for i in terms of v where

$$v = \frac{1}{1+i}.$$

Solution

Starting from

$$v = \frac{1}{1+i},$$

multiply both sides by $(1+i)$ to get

$$v(1+i) = 1.$$

Then

$$1 + i = \frac{1}{v}.$$

Subtracting 1 from both sides gives

$$i = \frac{1}{v} - 1.$$

Writing this as a single fraction,

$$i = \frac{1-v}{v}.$$

So the required formula is

$$\boxed{i = \frac{1-v}{v}}.$$